UNIVERSITY OF CALGARY
FACULTY OF SCIENCE
Department of Physics and Astronomy

# Simple Harmonic Oscillations

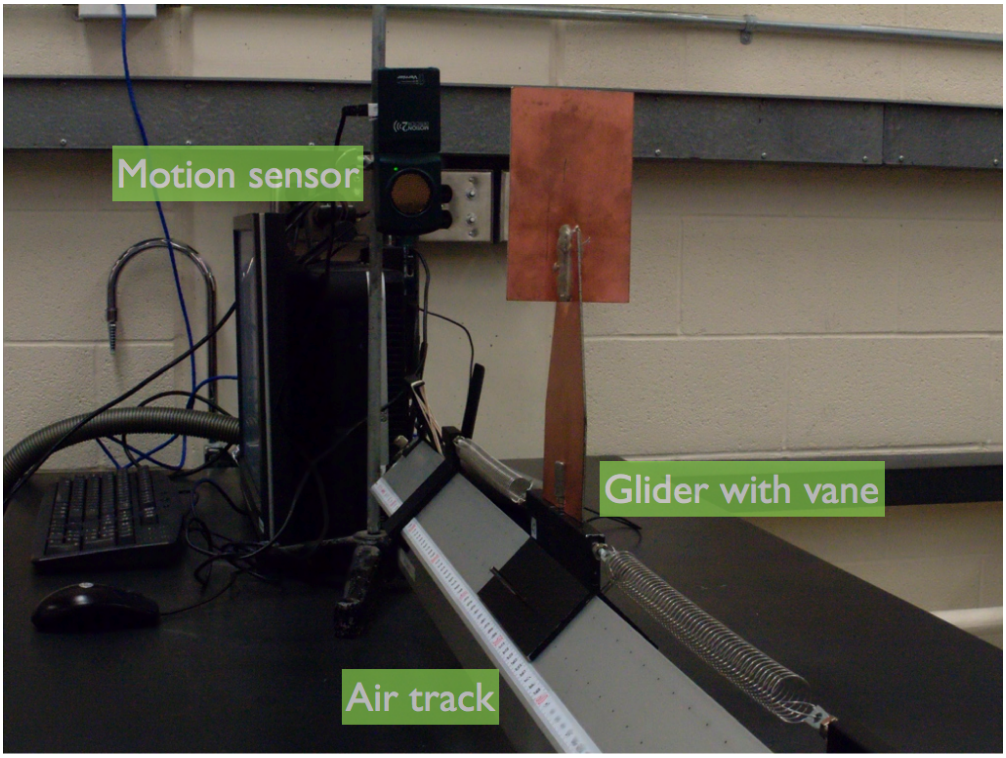



**Figure 1.1.** Simple harmonic oscillations experimental setup.

**Equipment:** Computer, air track, Pasco air supply, Pasco air hose, 20g gold mass [2], 50g silver mass [2], sail, long air table spring [2], banana plugs with eye hook [4], laboratory stand, right angle clamp, adjustable end stop, digital balance, Vernier position sensor, Vernier LabPro data logger, imperial threaded aluminum mounting rod.

**Goals of the Experiment:** To verify experimentally that a mass $m$ subject to an elastic restoring force with an elastic constant $k$ oscillates as a sinusoidal function of time, with angular frequency $\omega = \sqrt{k/m}$; Use relationships between angular frequency, mass and spring constant ($\omega = \sqrt{\frac{k}{m}}$), as well as acceleration and position $\left(a(t) = -\omega^2 x(t)\right)$ to determine the value of $m$.

# Theory

## Background

A mass oscillating on the end of a spring is the prototypical simple harmonic oscillator. In this experiment, the mass is a glider moving horizontally along an air track. The weight of the glider is thus cancelled by the reaction force exerted by the air track on the glider, and friction is made negligible by the cushion of air on which the glider moves.

Figure 1.2 shows the setup schematically. The glider is held between two identical springs of elastic constant $k$. At equilibrium, the springs exert equal and opposite forces on the stationary glider. If the glider is displaced a distance $\Delta x$ to the right the force exerted by spring on the right decreases by an amount $k\Delta x$, while the force exerted by the spring on the left increases by the same amount. The net force on the glider in the $+x$ direction is therefore:

$$F_x = -2k\Delta x = -k_{\text{effective}}\Delta x \tag{1.1}$$

where

$$k_{\text{effective}} = k_1 + k_2 \tag{1.2}$$

The glider moves with simple harmonic motion, with an angular frequency $\omega = \sqrt{2k/m}$.

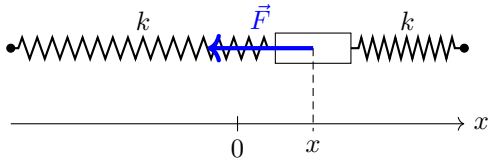



**Figure 1.2.** Schematic diagram of the setup: The glider between two springs.

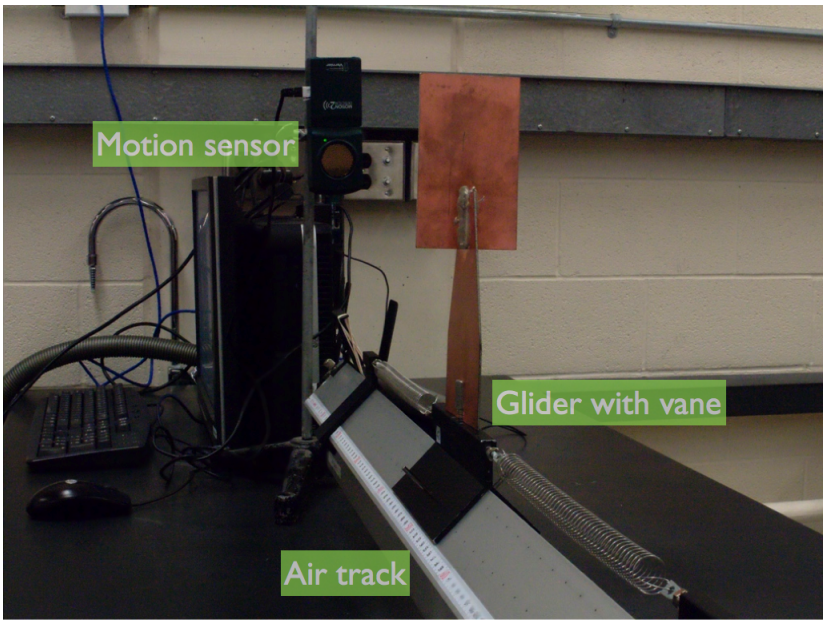



**Figure 1.3.** The experimental setup.

# The Experiment

## Overview of Experiment: The Setup

Figure 1.3 shows the actual setup. The glider has a vane attached, that can be tracked by the motion sensor, which bounces ultrasound waves off the vane, and measures distance through the travel time of the wave.

## Pre-Lab

**Your Pre-lab exercises are available on the D2L Lab site (Assessment – Quizzes) and should be completed there.**

## Experimental Procedure

1. Measure the mass $m$ of the glider using the electronic scale provided. Record the value $m$ with its uncertainty $u(m)$.
2. Get the Logger Pro software ready to collect data from the motion sensor. Turn on the air supply to the air track and set the glider in motion. Take approximate measurements of the period of the oscillation to make sure you can record at least a few full cycles. Collect data with Logger Pro.
3. Use the "Examine" tool to determine the period of the oscillation. Record the data sampling rate. Use the data sampling rate to estimate a Type B uncertainty for your period.
4. Select an interval (preferably as long as possible) that consists purely of simple harmonic motion on the plot of position vs. time. Use the "Analysis" tool to fit it with a curve of a form $A + B\cos(Ct + D)$.
5. Determine and record values of parameters $A$, $B$, $C$, and $D$ five times and find the uncertainty in $C$.
6. CHECKPOINT: Compare the values of the period determined using the "Examine" tool to that calculated from fitting a cosine curve to the x(t) graph. If the values agree within uncertainty, continue with the experiment. If not, re-examine your measurements/analysis.
7. Plot the acceleration as a function of the displacement. Fit acceleration versus displacement with a straight line. Record the values of slope and y-intercept with uncertainties.
8. Add additional mass $m'$ to the glider and set the glider in motion.Use the "Analysis" tool to fit it with a curve of a form$A + B\cos(Ct + D)$. Determine the angular frequency of the oscillation using x(t) graph and the slope and y-intercept of the a(x) graph. Make sure to record uncertainties for all the values. This data will be used to calculate the additional mass $m'$.
9. Measure the mass $m'$ using the scale provided. Record the value with its uncertainty.

# Data Analysis

1. Determine the effective spring constant $k_{\text{effective}}$ and the spring constant of each spring $k$ using information obtained from analyzing the x(t) graph. Using the standard deviation, compute a Type A uncertainty for $k$. Additionally, the measurement hardware has an inherent uncertainty in time of roughly 0.01 seconds. Use $u(\omega)/\omega = u(t)/t$, where $t$ is the total length of time of the measurement (i.e. several seconds) to estimate $u(\omega)$. Propagate that uncertainty to a Type B uncertainty for $k$. When you estimate an uncertainty using two methods, you can either combine them in quadrature or if one of the values dominates (i.e. is much larger than the other), you can use the larger value of the estimated uncertainly.

2. Determine the effective spring constant $k_{\text{effective}}$ and spring constant of each spring $k$ using information obtained from analyzing the a(x) graph. The uncertainty in the slope reported by Logger Pro is a Type A uncertainty (derived from repeated measurements). Estimate a Type B uncertainty using the same arguments as in the previous data analysis step. Report your values with uncertainties.

3. Compare these two values of the effective spring constant. Comment on their agreement within uncertainty.

4. Using the value ofthe spring constant obtained from the first part of the experiment (step 1), determine the value of the additional mass $m'$. Comment on any assumptions, estimates and choices.

5. Compare the values of the mass $m'$ obtained from the analysis of the simple harmonic oscillator and from the balance measurement. Comment on any discrepancies.

# Lab Report

Your lab report should include the following items:

- A title; your name, and those of your collaborators;
- Objective of the experiment;
- Short description of experimental methods;
- Data (with uncertainties):
  - the mass of the glider with its uncertainty;
  - the amplitudes, angular frequencies, initial phases, and zero offsets of the oscillation for both experiments;
  - the slopes and y-intercepts of a(x) graphs for both experiments;
  - the value of additional mass with its uncertainty obtained from the balance;
- Analysis:
  - calculations of the spring constant $k$ and its uncertainty using data from x(t) and a(x) graphs;
  - calculations of the additional mass $m'$;
  - calculations of all uncertainties;
- Discussion:
  - comparison of the values of $k$ and $m'$ within their uncertainties;
  - comments on validity of the simple harmonic oscillation as a method of determining the mass of the object;
  - comments on possible sources of uncertainties, including classifications of experimental errors as random or systematic;
- Conclusion about whether the values agree within uncertainties.